Dhruv Prajapati

Rochester, NY | +1 (520) 390-4462
ddp8909@rit.edu | linkedin.com/in/dhruv-prajapati1220 | dhruv-prajapati1220.vercel.app

Fourth-year **Mechanical Engineering** student specializing in **thermal-fluid systems**, **CFD**, and **failure analysis**, seeking a **Summer 2027** internship.

Education

Rochester Institute of Technology **Aug. 2023 – Dec. 2027**
Bachelor of Science in Mechanical Engineering, Senior Year, Dean's List 2026, GPA 3.50 *Rochester, NY*

Experience

Battery Cell Engineer Internship **Sept. 2026 – Dec. 2026**
Tesla *Austin, TX*

- Incoming intern working with 4680 tabless cell mechanical modeling & design, manufacturing, and process development teams to support cell development for Robotaxi/Cybercab and Cybertruck platforms.

Failure Analysis Engineer Internship **Jun. 2026 – Aug. 2026**
Pacific Gas & Electric Company, PG&E - ATS *San Ramon, CA*

- **Numerical Analysis/Simulation of Post Insulators** | *ANSYS Fluent + Maxwell + Structural, MATLAB*
- Developed a multiphysics transient simulation integrating CFD, electromagnetic and structural (FEA) analyses to evaluate 4 operational loading conditions and identify critical stress concentrations from gravity, aerodynamics (vortex shedding), and AC currents.
- Supported validation of 80 in. tall, L-type 230 kV Lapp porcelain insulators through mechanical testing, CFD post-processing, and technical reporting for engineering assessment.
- **Thermal Performance/Simulation Analysis** | *ANSYS Fluent, Experimental Test*
- Built a 2D ANSYS Fluent thermal model of a stranded aluminum conductor to evaluate heat flux and temperature response under high-temperature exposure and compared thermal behavior against rated operating limits.
- **Materials Characterization** | *SEM-EDS, Microscopy, MATLAB, Root Cause Analysis*
- Conducted microscopy investigations on more than 5 failed utility assets, including switches, conductors, and post insulators, to identify fracture mechanisms and support root-cause analysis.

Mechanical Engineer Researcher **Feb. 2026 – May 2026**
Fluids Acoustics Structures Laboratory, FAST LAB *Rochester, NY*

- **Test Platform Design** | *Arduino, Pololu Jrk Motor Controller, SolidWorks, 3D Printing*
- Designed and built an experimental vortex-ring test platform integrating a linear actuator, Arduino-based control system, and Pololu motor controller to investigate confined-flow fluid dynamics.
- **Vortex-Ring Impingement Heat Transfer** | *Particle Image Velocimetry (PIV), MATLAB*
- Conducted PIV experiments to quantify vortex-induced flow structures and impingement heat-transfer behavior in confined geometries.

Mechanical Test Engineer Internship **Aug. 2025 – Dec. 2025**
MKS Instruments *Rochester, NY*

- **Liquid-cooled Heat Sink Analysis** | *SolidWorks Flow Simulation, Experimental test bench*
- Designed and executed experimental characterization of a liquid-cooled microchannel heat sink for a 5 kW pulsed-DC generator (RPG-50), quantifying pressure-drop and flow-rate tradeoffs.
- Validated results with first principles and used CFD to identify flow maldistribution, enabling a 25% reduction in hydraulic pressure drop.
- **Cross-functional projects** | *ALT, Engineering Analysis, LabVIEW*
- Supported instrumentation and validation activities across multiple reliability test environments, including HVAC controls, liquid-nitrogen systems, and ALT/HASS/ESS chambers used for hardware qualification.

Technical Skills

Thermal & Fluid Systems: Heat Transfer, Fluid Mechanics, CFD (SolidWorks Flow Simulation, ANSYS Fluent)
CAD Design & Analysis: SolidWorks, ANSYS Mechanical & Fluent, CATIA v5
Programming: Python, MATLAB, Simulink, LabVIEW, Arduino
Hardware & Testing: Hitachi SEM, Oscilloscopes, Keysight Waveform generator, PIV, Instron
Manufacturing: CNC Milling, Lathe, GD&T, 3D Printing, Surface Metrology